

POMs-GibbsNet: A Deep Neural Network for the Prediction of Thermodynamic Parameters of Polyoxometalates

Jordi Buils¹, Carles Bo², David Balcells³

1. 1Institute of Chemical Research of Catalonia (ICIQ Barcelona Institute of Science and Technology (BIST

2. Departament de Química Física i Inorgànica Universitat Rovira i Virgili

3. Hylleras Centre for Quantum Molecular Sciences, Department of Chemistry University of Oslo

Abstract

Polyoxometalates (POMs) are versatile metal–oxo clusters with remarkable structural diversity and applications spanning catalysis, energy storage, and materials science. However, accurately predicting their thermodynamic properties remains computationally demanding due to the large system sizes and the cost of density functional theory (DFT) frequency calculations required to obtain Gibbs free energies. In this work, we present POMs-GibbsNet, a deep neural network designed to predict the thermal free energy correction (G–E) of polyoxometalates by combining graph-based molecular representations with DFT-derived descriptors. Molecular structures were encoded using variants of the AABBA autocorrelation graph kernel, which captures atom–atom, bond–bond, and atom–bond correlations together with quantum-derived features including QTAIM descriptors. POMs-GibbsNet was trained on a dataset of 1625 POM structures spanning 19 chemical systems. A multilinear regression model based on global molecular descriptors was first evaluated as a baseline, achieving R² values of 0.84–0.92 but with relatively large prediction errors (RMSE 9 kcal mol^{–1}). In contrast, POMs-GibbsNet captures nonlinear structure–property relationships and significantly improves predictive accuracy. External validation using POMSimulator reaction networks comprising over 9×10⁴ reactions shows excellent agreement between predicted and DFT reaction energies (R² > 0.99), with errors typically within a few kcal mol^{–1}. Notably, simplified atom–atom autocorrelation descriptors already provide robust performance while reducing model complexity. These results demonstrate that POMs-GibbsNet enables efficient estimation of thermodynamic corrections in POM systems, providing a powerful framework for accelerating the computational exploration of polyoxometalate speciation and self-assembly across large chemical spaces.

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Jordi Buils^{1,*}, Carles Bo^{1,2}, and David Balcells^{3,*}

¹Institute of Chemical Research of Catalonia (ICIQ), The Barcelona Institute of Science and Technology (BIST), Av. Països Catalans, 16, 43007 Tarragona, Spain.

²Departament de Química Física i Inorgànica, Universitat Rovira i Virgili, Marcel·lí Domingo s/n, 43007, Tarragona, Spain.

³Hylleras Centre for Quantum Molecular Sciences, Department of Chemistry, University of Oslo, 0315 Oslo, Norway

*Corresponding author

April 6, 2026

Abstract

Polyoxometalates (POMs) are versatile metal–oxo clusters with remarkable structural diversity and applications spanning catalysis, energy storage, and materials science. However, accurately predicting their thermodynamic properties remains computationally demanding due to the large system sizes and the cost of density functional theory (DFT) frequency calculations required to obtain Gibbs free energies. In this work, we present POMs-GibbsNet, a deep neural network designed to predict the thermal free energy correction ($G - E$) of polyoxometalates by combining graph-based molecular representations with DFT-derived descriptors. Molecular structures were encoded using variants of the AABBA autocorrelation graph kernel, which captures atom–atom, bond–bond, and atom–bond correlations together with quantum-derived features including QTAIM descriptors. POMs-GibbsNet was trained on a dataset of 1625 POM structures spanning 19 chemical systems. A multilinear regression model based on global molecular descriptors was first evaluated as a baseline, achieving R^2 values of 0.84–0.92 but with relatively large prediction errors (RMSE 9 kcal mol^{−1}). In contrast, POMs-GibbsNet captures nonlinear structure–property relationships and significantly improves predictive accuracy. External validation using POMSimulator reaction networks comprising over 9×10^4 reactions shows excellent agreement between predicted and DFT reaction energies ($R^2 > 0.99$), with errors typically within a few kcal mol^{−1}. Notably, simplified atom–atom autocorrelation descriptors already provide robust performance while reducing model complexity. These results demonstrate that POMs-GibbsNet enables efficient estimation of thermodynamic corrections in POM systems, providing a powerful framework for accelerating the computational exploration of polyoxometalate speciation and self-assembly across large chemical spaces.

1. Introduction

Polyoxometalates represent a remarkably versatile class of molecular metal–oxo clusters, comprising early transition metals in high oxidation states (typically Mo^{6+} , W^{6+} , V^{5+} , Nb^{5+} , Ta^{5+})^{1,2} interconnected by oxo ligands into discrete nanoscale architectures spanning 1 to 6 nm. First systematically explored in the late 19th century by Berzelius and elaborated through pioneering works in the 20th century—such as Keggin’s structural elucidation in 1933—these polynuclear anions have surged in relevance over the past decades due to their unprecedented structural diversity.^{2–5} This encompasses iconic archetypes like Keggin [$XM_{12}O_{40}$] ^{$n-$} , Dawson [$X_2M_{18}O_{62}$] ^{$m-$} , Anderson–Evans [XM_6O_{24}] ^{$p-$} , and Lindqvist [M_6O_{19}] ^{$q-$} , alongside lacunary derivatives (with metal vacancies for functionalization),^{4,6–8} transition-metal-substituted variants (e.g., lacunary Keggin with Co^{2+} , Cu^{2+} , Fe^{3+}), organo-functionalized hybrids, and even giant supramolecular assemblies exceeding 100 metal centers.^{6,9,10}

Their pH-responsive speciation, dynamic self-assembly pathways, and rich multi-electron redox chemistry in aqueous media—spanning multiple accessible potentials—enable transformative applications across homogeneous and heterogeneous catalysis^{11–13} (e.g., selective oxidations of alkanes, alkenes, and sulfides using $PW_{12}O_{40}^{3-}$), energy storage^{12,14}

(pseudocapacitors with capacitance > 500 F g^{−1}, lithium-ion battery cathodes), medicine^{15–18} (antiviral agents against HIV, influenza, and COVID-19; anticancer via ROS (reactive oxygen species) generation and DNA binding), photovoltaics, and advanced functional materials^{4,7,19} such as luminescent probes and sensors.

Yet, the intricate mechanisms governing their assembly—from mononuclear precursors via condensation to thermodynamically favored clusters—their inherent conformational flexibility (e.g., Jahn–Teller distortions in vanadium POMs), and pronounced sensitivity to counterions (e.g., alkali cations templating speciation), pH, and solvent effects pose formidable barriers to both experimental synthesis (low yields, purification challenges)^{9,20,21} and traditional computational modeling^{1,17,22} (e.g., DFT’s prohibitive scaling for > 100 -atom systems, inadequate solvation models, and sampling of protonation or tautomer states).²³ Accurately predicting key equilibrium constants—such as formation constants K_f , acid dissociation constants pK_a , and protonation sequences—is essential for constructing speciation diagrams, optimizing synthetic conditions, and elucidating reactivity.^{21,24,25} However, this remains notoriously challenging due to sparse experimental data and modeling inaccuracies, underscoring an urgent need for innovative predictive tools such as machine learning.²⁶

POMSimulator²⁵ represents a pioneering computational

platform tailored for modeling the speciation, self-assembly pathways, and thermodynamic stability of polyoxometalates under varying pH, counterion, and solvent conditions. By integrating density functional theory calculations with statistical post-processing,^{21,24} it circumvents the prohibitive computational costs of explicit high-level quantum modeling for large clusters, enabling the construction of comprehensive Pourbaix diagrams and equilibrium speciation maps essential for synthetic design and reactivity prediction.

Recent advancements in POMSimulator feature a universal scaling²⁷ protocol that applies a linear correction to refine DFT-derived formation constants (K_f).

$$\log K_f^{\text{scaled}} = m \cdot \log K_f^{\text{DFT}} + b$$

Here, the slope m exhibits remarkable system-independence across diverse POM families, reflecting consistent systematic DFT biases in metal-oxo condensation energetics. The intercept b , however, is dynamically predicted through a multilinear regression model trained on an extensive array of statistical descriptors—such as mean absolute deviations, variance in bond orders, and cluster size metrics—extracted from unscaled DFT outputs. This hybrid approach mitigates errors from functional choice and solvation approximations while enabling reliable extrapolation to unexplored chemical space.

Building on these hybrid DFT-statistical methods exemplified by POMSimulator, machine learning emerges as a powerful paradigm to tackle longstanding challenges in polyoxometalate thermodynamics.²⁶ While DFT excels at providing mechanistic insights into metal-oxo bonding and condensation energetics, it struggles with scalability for clusters exceeding dozens of metal centers, inaccuracies in explicit solvation models beyond continuum approximations like COSMO, and the combinatorial explosion of configurational sampling required to account for protonation isomers, counterion pairings, and dynamic assembly pathways.²⁸ Machine learning circumvents these hurdles by constructing data-driven surrogate models that learn complex structure-property relationships from high-fidelity quantum datasets, delivering predictions orders of magnitude faster while enabling uncertainty quantification and active learning strategies.²⁹

Key to this advancement are molecular representations and architectures tailored for inorganic systems. Graph-based methods, such as message-passing neural networks or kernel encodings, effectively capture the topological intricacies of POMs, including the symmetries of the addenda metal-oxo framework metal (e.g., Keggin α/β isomers), lacunary defects, and polyhedral connectivity. Models trained on specialized datasets such as tmQM³⁰ or Hessian QM9³¹ facilitate accurate inference of vibrational free energies, protonation constants (pK_a), formation equilibria (K_f), and redox potentials.

For POMs specifically, machine learning bridges the gap between computationally demanding DFT pipelines and experimental imperatives, as demonstrated by POMSimulator’s statistical post-processing for scaling formation constants. These approaches integrate seamlessly with global optimization techniques, such as genetic algorithms or Bayesian optimization, to explore vast chemical spaces for stable POM

variants.

A critical aspect of applying machine learning to POMs is generating effective molecular fingerprints that capture their topological complexity, metal-oxo connectivity, and conformational flexibility. Common approaches include:

- **Traditional fingerprints:** Bit-vector methods such as Extended Connectivity Fingerprints (ECFP/Morgan)³² or Daylight fingerprints, which encode substructures but struggle with inorganic symmetries and multicenter bonds.
- **Descriptor-based methods:** Physicochemical descriptors from tools like RDKit or Dragon, aggregating atomic properties but often lacking graph topology.
- **Local environment descriptors:** Smooth Overlap of Atomic Positions or Atom-Centered Symmetry Functions, ideal for periodic solids but less suited for discrete clusters.
- **Graph neural networks:** Message-passing schemes (e.g., SchNet,³³ DimeNet^{34,35}) that learn iterative embeddings from node and edge features.

Among these approaches, graph-based kernels such as AABBA³⁶ excel for transition metal complexes, as well as for other clusters such as MOFs,³⁷ hinting at promising applications for POMs.

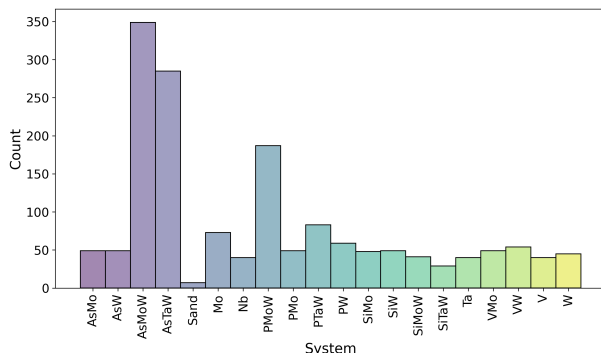


Figure 1: Bar plot for the number of polyoxometalate species in the dataset, grouped by system, with a global count of 1625 molecular graphs.

The AABBA graph kernel encodes molecular structures as fixed-length vectors through atom-atom, bond-bond, and atom-bond autocorrelations, systematically capturing both local chemical environments and global topological features essential for POMs. Unlike atom-only representations, AABBA incorporates bond order and path information, enabling superior performance in predicting energy barriers, bond distances, and thermodynamic properties. Its modular design also allows dimensionality reduction for interpretability, making it particularly suitable for integration with DFT features in models such as POMs-GibbsNet.

POMs-GibbsNet is a deep neural network designed to predict POM thermodynamic parameters by fusing DFT-derived features with advanced molecular representations. Trained on a curated dataset of 20 diverse POM systems, represented in Figure 1. From simple IPA^{21,38,39} systems: Mo,

W, V, Nb and Ta; typical HPAs: SiW, PW, PMo,²⁴ or AsMo; other kind of mixed poms: AsMoW, AsTaW, AsW, PMoW, PTaW, SiMo, SiMoW, SiTaW, VMo, VW, and Sandwich compounds (classified in this group because of their similar structure in which two Keggin lacunary structures form a sandwich-like structure POM-POM, or POM- M_3 -POM). This model addresses systematic DFT errors in stability and solvation energies. Integrating insights from POMSimulator and quantum datasets, POMs-GibbsNet provides a scalable framework for POM design in catalysis and materials science.

2. Methodology

2.1. DFT

All DFT calculations were performed using the SCM ADF 2019.14⁴⁰ package with the PBE^{41,42} functional and TZP basis set for all elements. Geometry optimizations and frequency calculations incorporated the Zero-Order Regular Approximation (ZORA)^{43,44} to account for relativistic effects. Water solvent effects were modeled using the COSMO solvation model implemented in ADF with Klamt’s default radii.⁴⁵ Stationary points were characterized through analytical frequency analysis, yielding ground-state Gibbs free energies at 1 atm and 298.15 K.

Quantum Theory of Atoms in Molecules (QTAIM)^{46–48} analysis was performed to rigorously define molecular connectivity and bonding. QTAIM, developed by Richard Bader, employs the topology of the electron density distribution to partition molecules into atomic basins separated by zero-flux surfaces and to identify bond critical points (BCPs) along bond paths.

At these BCPs, key topological parameters were extracted to characterize bond nature and strength in polyoxometalates. The electron density^{49,50} $\rho(\mathbf{r})$ quantifies electron accumulation at the BCP, with higher values indicating stronger bonding interactions. The Laplacian^{50–52} $\nabla^2\rho(\mathbf{r})$ distinguishes interaction types: negative values signify covalent shared interactions, while positive values denote closed-shell ionic or hydrogen bonding.

Bond ellipticity^{53,54} measures deviations from cylindrical symmetry and reveals π -character in bonds.

Abramov’s⁵⁵ energy densities at BCPs include the local kinetic energy density $G_b > 0$, potential energy density $V_b < 0$, and total energy density

$$H_b = G_b + V_b$$

where $H_b < 0$ indicates covalent character and correlates with bond dissociation energy.

Recent work has highlighted the growing relevance of QTAIM-derived descriptors in machine learning frameworks for complex chemical systems. In particular, high-throughput implementations of QTAIM enable the systematic generation of electron density-based features that can be directly integrated into graph neural network architectures for molecular and reaction property prediction.⁵⁶ These studies demonstrate that QTAIM features consistently improve predictive performance across diverse datasets, while also enhancing generalization to out-of-domain chemical spaces and low-data regimes. In parallel, multi-level QTAIM-enriched

graph neural networks have shown that incorporating ab initio topological descriptors across different levels of the theory leads to improved robustness and accuracy in predicting properties of transition metal complexes, particularly in extrapolative scenarios involving unseen compositions or limited training data.⁵⁷ Together, these advances reinforce the value of QTAIM as a physically grounded and transferable descriptor set for data-driven modeling, supporting its incorporation into the present framework.

2.2. AABBA

The AABBA graph kernel, as described by Morán et al.,³⁶ was employed to generate fixed-length vector representations of polyoxometalate molecular graphs. This kernel computes autocorrelation functions over atom–atom, bond–bond, and atom–bond pairs up to a defined path depth, encoding both local chemical environments and global structural motifs.

Polyoxometalates feature multiple metal centers and therefore require the “full” approximation. The maximum autocorrelation depth was set to half of the median maximum depth across the dataset. Three arithmetic operations (multiplication, sum and division) were tested independently to determine the highest-performing configuration.

These autocorrelation descriptors were augmented with node, edge, and whole-graph properties. Node features included atomic number, node degree, covalent radius, electronegativity, and DFT-derived multipole charges. Edge features incorporated QTAIM properties^{53,55,58,0} (ρ , $\nabla^2\rho$, ϵ , $\diamond$, metallicity indices, and Abramov energy densities G_b , V_b , H_b), together with bond distance and estimated bond order.

Whole-graph features included the total number of atoms and electrons, molecular charge, molecular weight, COSMO surface area and volume, HOMO and LUMO energies, HOMO–LUMO gap, electronic energy, and Gibbs free energy.

The graphs were labeled using the difference between Gibbs free energy G and electronic energy E :

$$\Delta G_{\text{thermal}} = G - E$$

2.3. POMs-GibbsNet Architecture and Training Hyperparameters

Hyperparameter optimization was performed using the Optuna⁵⁹ Python library, evaluating 100 configurations and selecting the best-performing model. Optimization was carried out using the full AABBA vector representation together with whole-graph properties using the multiplication as operator.

The resulting deep neural network architecture consisted of:

- An input layer with 126 neurons (AABBA + graph properties), 112 neurons (AABBA) or 33 neurons (AA).
- Two hidden layers with 53 and 27 nodes.
- A single-node output layer

Each layer employed an RELU activation function. Model training used the ADAM optimizer to minimize the mean

squared error loss. The dataset was split into training, validation, and test sets using a 70/15/15 ratio. Batch size was set to 20, and learning rate to 0.00688.

2.4. POMSimulator

POMSimulator is a computational framework specifically designed to address the challenging problem of polyoxometalate (POM) speciation and self-assembly in aqueous solution. Unlike conventional approaches that rely on manual construction of reaction schemes, POMSimulator automates the generation of extensive chemical reaction networks directly from density functional theory (DFT) data. By systematically identifying possible transformations between molecular species—including condensation, protonation, and addition reactions—the method constructs large-scale reaction networks that capture the complex multi-equilibria processes governing POM formation. This automated strategy enables the exploration of thousands of chemically plausible reactions, providing a comprehensive description of the underlying thermodynamic landscape.

A central feature of POMSimulator is its ability to translate these reaction networks into speciation models, which are subsequently solved to obtain concentration–pH diagrams and phase diagrams. The methodology relies on statistical sampling of a large number of speciation models, allowing the extraction of averaged predictions together with associated uncertainties. This approach has proven particularly powerful for systems such as phosphomolybdate clusters, where it successfully reproduces experimentally observed speciation behavior and reveals key intermediates involved in self-assembly pathways. In addition, the large-scale nature of the generated datasets enables the identification of dominant reaction pathways and building blocks, offering mechanistic insights that are difficult to access experimentally.

Despite its predictive capabilities, the accuracy of POMSimulator is inherently linked to the quality of the underlying DFT-derived thermodynamic data. Systematic errors in calculated formation constants, particularly those associated with solvation effects and entropy contributions, can propagate through the reaction network and affect speciation predictions. To address this limitation, recent developments have introduced a universal scaling approach that corrects DFT-derived formation constants through a linear transformation. In this framework, the slope of the scaling relationship is found to be largely system-independent, while the intercept can be predicted using a multilinear regression model trained on statistical descriptors of the dataset. This hybrid DFT–statistical methodology significantly improves the agreement between predicted and experimental equilibrium constants, enabling reliable predictions even for systems with limited experimental data.

Within this context, POMSimulator provides an ideal platform for validating machine learning models such as POMs-GibbsNet. By generating large and diverse reaction networks, it allows the assessment of predicted thermodynamic quantities across a wide range of chemical transformations and molecular systems. The integration of machine learning corrections into the POMSimulator workflow further enhances its efficiency by replacing computationally expensive

DFT-derived thermal contributions with rapid surrogate predictions. As a result, the combined POMSimulator–POMs-GibbsNet framework enables high-throughput exploration of polyoxometalate speciation and self-assembly processes, opening new avenues for the rational design and discovery of complex metal–oxo clusters.

3. Results and Discussion

3.1. MLR Baseline

As an initial reference point for evaluating the performance of more sophisticated machine learning models, a multilinear regression (MLR) model was constructed as a baseline predictor. This model employed exclusively whole-graph attributes describing global physicochemical properties of each polyoxometalate system. The selected descriptors included the total number of atoms and electrons, overall molecular charge, molecular weight, COSMO-derived solvent accessible surface area and molecular volume, and frontier orbital energies (HOMO and LUMO), together with the HOMO–LUMO energy gap obtained from DFT calculations. Furthermore, the electronic energy E and the Gibbs free energy G were used to construct the regression target $G - E$, corresponding to the thermal free energy correction.

These descriptors were selected because they provide a coarse-grained yet physically meaningful description of molecular size, electronic structure, and solvation-related effects, all of which are expected to influence thermodynamic quantities. In particular, electronic descriptors such as the HOMO–LUMO gap and molecular charge reflect electronic structure and reactivity trends, whereas geometric descriptors such as molecular surface area and volume capture size-dependent and solvation contributions to the free energy. The MLR model attempts to establish a linear relationship between these global features and the thermal correction term ($G - E$), which represents the vibrational, enthalpic, and entropic contributions to the Gibbs free energy.

The performance of the MLR model was evaluated using a standard train–test split of the dataset. In addition, to assess the influence of anomalous data points, an outlier filtering procedure was applied (using the IQR method), and the model was trained and evaluated both on the original dataset and on a reduced dataset with outliers removed. The resulting statistical metrics are summarized in Table 1. For the dataset including outliers, the model achieved a coefficient of determination of $R^2 = 0.84$ on the training set and $R^2 = 0.92$ on the test set, with corresponding RMSE values of 8.9 and 9.9 kcal mol^{−1}, and MAE values of 6.4 and 6.8 kcal mol^{−1}, respectively.

Upon removal of outliers, a noticeable improvement in error metrics was observed for both training and test sets, with RMSE decreasing to 6.3 and 6.5 kcal mol^{−1} and MAE to 4.9 and 5.0 kcal mol^{−1}, respectively. The R^2 values show a slight increase for the training set (0.87), while a modest decrease is observed for the test set (0.89), suggesting a reduction in variance driven by extreme data points. These results indicate that a subset of the largest prediction errors originates from atypical or difficult-to-model structures, and that the overall performance of the linear model is sensitive to their inclusion.

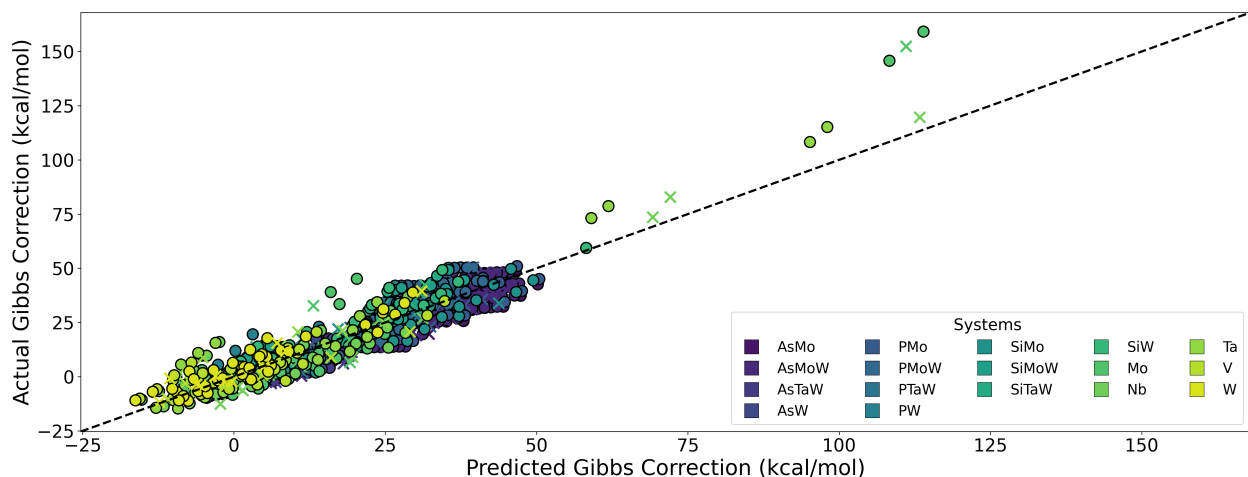


Figure 2: Multi-Linear Regression prediction vs DFT pair plot. Each datapoint is colored according to the polyoxometalate system they belong to. Each datapoint is represented as a circle or a cross depending if they belong to the training set or the test set respectively.

Table 1: Comparison of multilinear regression performance with and without outliers. Total samples consist of train and test.

Metric	With Outliers	Without Outliers
R^2 (Train)	0.84	0.87
R^2 (Test)	0.92	0.89
RMSE (Train)	8.9	6.3
RMSE (Test)	9.9	6.5
MAE (Train)	6.4	4.9
MAE (Test)	6.8	5.0
Total samples	1426	1372

Despite the reasonably strong correlations obtained with this approach, the magnitude of the prediction errors highlights the intrinsic limitations of purely linear models (as shown in Figure 2) for describing the thermodynamic behavior of polyoxometalate clusters. Even after outlier removal, errors on the order of $\sim 5\text{--}7\text{ kcal mol}^{-1}$ remain significant, particularly considering that thermal free energy corrections often span similar energy ranges across different cluster compositions and sizes. This indicates that a substantial portion of the variance in $G - E$ cannot be adequately captured through simple linear combinations of global molecular descriptors.

Such limitations are expected given the structural and electronic complexity of polyoxometalates. These clusters contain multiple metal centers connected through extended metal-oxo frameworks, often exhibiting subtle variations in bonding topology, electron delocalization, and coordination environments. As a consequence, nonlinear relationships between structural descriptors and thermodynamic quantities are likely to arise and cannot be effectively modeled within the constraints of the MLR framework.

Consequently, although the multilinear regression model provides a computationally inexpensive baseline, its predictive accuracy remains insufficient to reliably estimate thermal free energy corrections across chemically diverse polyoxometalate systems. To address these limitations and capture the nonlinear structure-property relationships inherent to these clusters, a deep neural network architecture,

termed POMs-GibbsNet, was subsequently developed and evaluated.

3.2. POMs-GibbsNet Performance

To evaluate the effectiveness of the POMs-GibbsNet architecture and determine the most suitable molecular representation for the prediction task, several variants of the AABBA autocorrelation vector were systematically tested. These representations differ in their level of structural detail and dimensionality, allowing an assessment of how different types of chemical information contribute to predictive accuracy.

The most comprehensive representation consisted of the full AABBA graph kernel augmented with whole-graph descriptors, resulting in a feature vector containing 126 attributes. The use of the complete AABBA kernel was motivated by previous work by Morán *et al.*, which demonstrated that this representation performs particularly well for transition metal complexes by capturing both local chemical environments and long-range structural correlations within the molecular graph. By combining atom-atom, bond-bond, and atom-bond autocorrelations, the AABBA representation provides a rich description of the connectivity patterns and bonding environments characteristic of polyoxometalate clusters.

Within this feature space, three arithmetic operations were explored for constructing the autocorrelation descriptors: multiplication, division, and summation. These operations control how the atomic and bond properties propagate along paths in the molecular graph, thereby affecting the sensitivity of the representation to different structural motifs. Although the performance according to statistical metrics (R^2 , RMSE and MAE) was quite similar for the three configurations, multiplication was slightly better and therefore selected for subsequent hyperparameter optimization and final model training (Figure 3).

In order to investigate whether a reduced feature space could maintain predictive accuracy while improving computational efficiency and interpretability, the representation was subsequently simplified by removing the additional whole-

graph descriptors. This reduced configuration consisted solely of the full AABBA kernel (112 features), incorporating Atom–Atom, Bond–Bond, and Atom–Bond autocorrelations. As in the previous case, the three mathematical operations (multiplication, division, and sum) were evaluated independently.

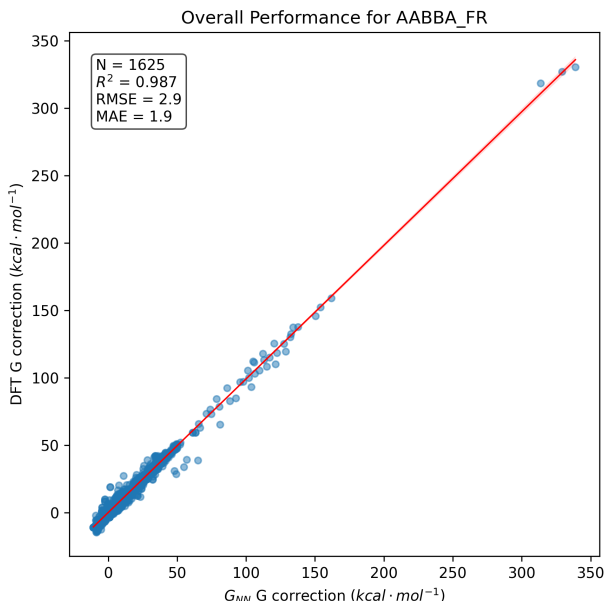


Figure 3: POMs-GibbsNet predictions vs DFT pair plots using AABBA autocorrelation vector plus whole graph features using the product operation.

The resulting models displayed performance trends similar to those observed for the larger representation. Like before, sum and multiplication operations perform similarly and clearly better than division.

Finally, a minimal baseline representation was evaluated in order to determine the extent to which predictive performance depends on the inclusion of bond-level information. In this simplified scenario, only Atom–Atom autocorrelations were used, producing a 33-feature vector that was combined with the three mathematical operations. Despite the drastic reduction in feature dimensionality, the resulting models still achieved comparable performance levels, indicating that even relatively simple graph-based descriptors can capture important aspects of the structural variability within the dataset. A quantitative comparison of the different representations is provided in Table 2. Among these minimal representations, the sum operation once again provided marginally better predictive performance, although the differences between operations were relatively small.

Taken together, these experiments demonstrate that POMs-GibbsNet is robust with respect to variations in feature representation and dimensionality. While richer feature sets incorporating bond-level information and QTAIM descriptors generally provide slightly improved accuracy $R^2 > 0.99$, even reduced representations retain significant predictive capability $R^2 \sim 0.99$ and $MAE < 2.2 \text{ kcal} \cdot \text{mol}^{-1}$. This robustness suggests that the learned relationships between graph features and thermodynamic corrections are physically meaningful and transferable across different polyoxometalate systems.

Table 2: Performance of different autocorrelation (AC) representations and operations after model optimization.

AC Type	Operation	N	R^2	RMSE	MAE
AA	Multiplication	1625	0.989	2.6	1.7
AA	Division	1625	0.985	3.1	2.2
AA	Sum	1625	0.993	2.1	1.5
AABBA	Multiplication	1625	0.995	1.7	1.3
AABBA	Division	1625	0.987	2.9	1.9
AABBA	Sum	1625	0.996	1.6	1.1
Ext. AABBA	Multiplication	1625	0.997	1.3	0.8
Ext. AABBA	Division	1625	0.993	2.0	1.4
Ext. AABBA	Sum	1625	0.997	1.4	1.0

3.3. Application in automated chemical reaction network: reaction Gibbs free energies.

To further assess the reliability and practical applicability of the machine learning predictions, the POMs-GibbsNet results were validated using the POMSimulator platform. POMSimulator is capable of automatically generating extensive reaction networks for polyoxometalate systems, including thousands of possible chemical transformations derived from combinations of cluster building blocks, protonation states, and condensation pathways. This capability makes it particularly suitable for evaluating the consistency of predicted thermodynamic quantities across chemically diverse reactions. After these reaction networks are generated, POMSimulator calculates the reaction energy using the absolute Gibbs free energies of the molecules involved in the reaction. With these reaction energies, POMSimulator is able to simulate the aqueous multi-species equilibria of polyoxometalates and then compute the corresponding formation constants.

Within this validation framework, the Gibbs free energies predicted by the neural network were reconstructed by adding the predicted thermal correction term to the electronic energies obtained from the DFT calculations. In contrast, the reference values were taken directly from the Gibbs free energies computed using full DFT frequency calculations. By comparing these two quantities, it becomes possible to assess whether the machine learning model accurately reproduces the thermodynamic contributions arising from vibrational and entropic effects.

In opposition to typical ML energy prediction in which absolute energies are used, our approach relies on the validation of a more realistic metric, which are reaction energies. With this validation methodology, we can assess whether the error in our NN predictions accumulate worsening the results, or if they actually cancel each other in analogy to DFT.

In this validation procedure, different reaction types were considered, including condensation, acid–base, addition, and hydration reactions. These reactions represent the fundamental transformations that govern the self-assembly and speciation of polyoxometalate in an aqueous solution. For each reaction type, reaction energies were calculated using both the DNN-corrected Gibbs free energies and the original DFT-derived Gibbs energies. Linear regression analyses were subsequently performed to quantify the agreement between the predicted and reference reaction energies.

The statistical results for each reaction category are summarized in Table 3, which reports the number of reactions analyzed together with the corresponding R^2 , RMSE, and MAE

values for each reaction class. In all cases, the predicted reaction energies exhibit extremely strong linear correlations with the DFT values, confirming that the machine learning corrections successfully reproduce the thermodynamic trends across the reaction network. Acid-base reactions, which represent the largest portion of the dataset, display particularly small prediction errors, with root mean squared errors close to 1 kcal·mol⁻¹. Addition and condensation reactions show slightly larger deviations, although their correlations remain very strong, with R^2 values exceeding 0.99. Hydration reactions, which constitute a very small fraction of the dataset, exhibit somewhat larger uncertainties due to the limited number of available examples. Nevertheless, the overall agreement remains excellent, demonstrating that the POMs-GibbsNet predictions are sufficiently accurate to reproduce the thermodynamic landscape generated by full DFT calculations.

Table 3: Statistics for the linear regressions between NN-derived and DFT-computed Gibbs free energies, grouped by reaction type. Only the extended AABBA vector using the multiplication was considered for this study (highest R^2 and lowest RMSE and MAE), giving a total of 10819 reactions. RMSE and MAE are in kcal/mol.

Reaction type	N	R^2	RMSE	MAE
Acid-Base	6944	0.9999	1.1	0.7
Hydration	6	0.9998	2.3	1.8
Addition	1668	0.9977	2.1	1.6
Condensation	2201	0.9960	2.2	1.9

To gain further insight into the behavior of the model, the reaction dataset was analyzed from two complementary perspectives: according to the reaction type, and according to parent polyoxometalate system involved in the reaction. This multi-level analysis allows the identification of potential systematic trends in the predictive performance and provides insight into how structural complexity influences the thermodynamic corrections learned by the neural network.

First, the reactions were grouped according to the autocorrelation representation employed during model training. In general and in line with the results for absolute energies (Table 2), models incorporating the extended AABBA kernel produced slightly improved agreement with DFT reaction energies, particularly for transformations involving significant structural rearrangements, such as condensation and addition reactions. This behavior suggests that bond-level descriptors and QTAIM-derived features capture relevant electronic and topological changes occurring along these reaction pathways. Conversely, simplified atom-atom representations still produced reasonable predictions but occasionally exhibited larger deviations for reactions involving subtle bond reorganization. Notably, AA autocorrelations already provide good results for the POMSimulator framework (e.g., $R^2 \approx 0.999$, RMSE $\approx$ 2-3 kcal mol⁻¹; Table 4), helping to avoid overfitting while maintaining high predictive accuracy.

Second, the reaction energies were analyzed by polyoxometalate system in order to determine whether predictive accuracy varies across different metal compositions and structural families. The corresponding statistical metrics are reported in Table 5. The dataset contains a wide variety of POM chemistries, including tungsten-, molybdenum-,

Table 4: Performance by autocorrelation representation. Models are grouped by descriptor type. Ext. AABBA vectors consist of AABBA autocorrelation features augmented with whole-graph properties. Each model consisted of 10819 reactions. RMSE and MAE are reported in kcal mol⁻¹.

Model	Operator	R^2	RMSE	MAE
AA	Sum	0.9995	2.4	1.6
AA	Division	0.9992	3.2	2.2
AA	Multiplication	0.9991	3.2	2.1
AABBA	Sum	0.9998	1.7	1.1
AABBA	Multiplication	0.9997	1.8	1.2
AABBA	Division	0.9992	3.1	2.0
Ext. AABBA	Sum	0.9998	1.6	1.0
Ext. AABBA	Multiplication	0.9998	1.6	1.1
Ext. AABBA	Division	0.9996	2.3	1.5

niobium-, vanadium-, and mixed-metal clusters. Grouping reactions according to their parent system enables identification of potential systematic biases associated with specific metal compositions or structural motifs. Overall, the results revealed consistent predictive performance across most POM families, indicating that the POMs-GibbsNet model successfully generalizes across chemically diverse polyoxometalate architectures, even though some trends could be observed.

Mixed-metal POM families exhibited lower error values compared to classical IPA POMs. This difference does not pose a concern, as classical polyoxometalates have been widely studied with DFT, whereas mixed-metal POMs have received less attention, providing an opportunity to enhance understanding of these more complex systems, where the huge amount of possible combinations arising from substitutions in the POM scaffold directly prompts for the development of faster calculation strategies like ours.

Table 5: Prediction performance by POM system for the extended AABBA representation. RMSE and MAE are reported in kcal mol⁻¹.

System	$N_{\text{reactions}}$	R^2	RMSE	MAE
AsTaW	2364	1.000	1.1	0.7
AsMoW	3532	1.000	1.0	0.8
SiTaW	19	0.999	1.3	1.0
AsW	204	0.998	1.6	1.3
PTaW	226	0.998	2.6	2.0
PMoW	1931	0.998	2.3	1.6
SiW	200	0.998	2.3	1.9
VW	252	0.997	1.8	1.4
Mo	255	0.997	2.8	2.0
PW	309	0.997	2.2	1.6
SiMo	180	0.997	2.6	2.1
Nb	118	0.996	2.5	2.0
SiMoW	38	0.995	2.1	1.3
Ta	118	0.995	2.5	2.1
AsMo	204	0.995	2.3	1.8
V	224	0.993	2.6	2.1
VMo	204	0.991	2.7	2.1
PMo	204	0.991	2.8	2.2
W	237	0.977	3.4	2.5

These observations highlight the importance of maintaining chemical diversity in the training dataset to ensure the transferability of the learned thermodynamic corrections.

Taken together, these analyses demonstrate that the machine learning predictions remain robust across different

molecular representations, polyoxometalate systems, and reaction classes. The consistent agreement between predicted and DFT-derived reaction energies highlights the reliability of the proposed DFT-machine learning hybrid framework. By accurately reproducing the thermodynamic corrections required to obtain Gibbs free energies, the POMs-GibbsNet model enables rapid evaluation of extensive reaction networks while avoiding the computational cost associated with full vibrational frequency calculations. When integrated with tools such as POMSimulator, this approach provides a powerful strategy for accelerating the exploration of polyoxometalate speciation and self-assembly pathways across large regions of chemical space.

4. Data Records

4.1. POMs repository: *ioChem-BD*

A total of 1625 polyoxometalates, sorted into 19 different systems, were used to perform the AABBA autocorrelations, to create a fixed-length vector that could be fed into different ML algorithms. To check whether these polyoxometalates had an heterogeneous distribution of their properties, density plots were employed for each of the graph properties (Figure SI I).

Additionally, bar plots were used to study the maximum depth of graphs. The depth can be defined as the maximum distance between two nodes of the graph. This step was important for the generation of the AA, BB and AB autocorrelation vectors. The median value of this distribution was used to proceed with the ACs.

The resulting dataset was uploaded at the *ioChem-BD* repository⁶⁰ and can be publicly accessed via the following link <https://doi.org/10.19061/iochem-bd-1-423>.

4.2. Code availability: *GitLab*

An open repository of the code is available in *GitLab* (<https://gitlab.com/JBuils/poms-gibbsnet>). Additionally to the POMs-GibbsNet model, the repository contains an adapted version of the AABBA graph kernel and an adapted version of the POMSimulator 1.0.0 release.

5. Conclusions

In this work, a hybrid DFT and machine learning framework was developed to efficiently predict the thermal free energy corrections of polyoxometalate clusters. The proposed model, POMs-GibbsNet, combines graph-based molecular representations with DFT-derived descriptors to learn the relationship between structural features and the thermal contribution to the Gibbs free energy ($G - E$). By focusing on this correction term rather than absolute energies, the model isolates the vibrational, enthalpic, and entropic contributions that are typically obtained from computationally expensive frequency calculations.

As an initial benchmark, a multilinear regression model based solely on global molecular descriptors was evaluated. Although the MLR approach captured broad trends in the dataset, prediction errors remained relatively large, demonstrating the limitations of linear models for describing the

complex structural and electronic effects present in polyoxometalate systems. These observations motivated the development of the POMs-GibbsNet deep neural network architecture, which can capture nonlinear relationships between molecular graph features and thermodynamic corrections.

Different graph-based molecular representations were explored using variants of the AABBA autocorrelation kernel. The results demonstrate that descriptors incorporating atom-atom, bond-bond, and atom-bond correlations provide a rich description of the metal-oxo frameworks characteristic of polyoxometalates. In particular, the inclusion of QTAIM-derived bond descriptors improved the ability of the model to capture subtle electronic effects associated with metal-oxo bonding. Nevertheless, simplified atom-atom autocorrelation representations also produced highly competitive predictive performance despite their significantly lower dimensionality. This observation suggests that much of the relevant structural information governing thermal corrections is already encoded in the global atomic connectivity patterns of the clusters. From a practical standpoint, the use of reduced AA descriptors offers an attractive compromise between model simplicity and predictive accuracy, while also helping to mitigate potential overfitting.

The reliability of the POMs-GibbsNet predictions was further assessed through external validation using the POMSimulator platform. Extensive reaction networks comprising condensation, acid-base, addition, and hydrolysis reactions were generated and used to compare reaction energies obtained from DFT-derived Gibbs free energies with those reconstructed using the machine learning corrections. Across all reaction classes, the predicted energies exhibited excellent agreement with the reference DFT values, with coefficients of determination exceeding 0.99 and root mean squared errors typically within a few kcal mol⁻¹. Additional analyses grouped by molecular representation and polyoxometalate system confirmed that the model maintains robust predictive performance across a wide range of metal compositions and structural families.

Overall, these results demonstrate that POMs-GibbsNet provides a reliable and computationally efficient strategy for approximating thermal free energy corrections in polyoxometalate systems. By eliminating the need for explicit vibrational frequency calculations, this approach dramatically reduces computational cost—from days using DFT to minutes with POMs-GibbsNet—enabling the efficient evaluation of large sets of candidate structures. When integrated with automated reaction network generators such as POMSimulator, the proposed framework enables rapid exploration of polyoxometalate self-assembly pathways and speciation landscapes across broad regions of chemical space.

More broadly, the combination of graph-based molecular descriptors, physically informed DFT features, and machine learning models represents a promising direction for accelerating the study of complex inorganic clusters. The methodology presented here could be extended to other classes of metal-oxo systems, coordination complexes, or catalytic clusters, providing a general strategy for bridging the gap between accurate quantum chemical calculations and large-scale chemical discovery.

Author contributions

JB adapted the AABBA graph kernel, developed and implemented the POMs-GibbsNet neural network model, and adapted POMSimulator. JB also optimized and tested the model's performance, generated the POMs database, and uploaded it to ioChem-BD. JB was the main contributor to writing the manuscript. CB and DB contributed to the design and supervision of the research project, revised the manuscript, and secured funding. All authors contributed to writing and revising the manuscript.

Acknowledgements

J. B. and C. B. acknowledge the Spanish Ministry of Science, Innovation and Universities MCIN/AEI/10.13039/501100011033 (PID2023-153344NB-I00, and CEX2024-001469-S), the ICIQ Foundation and the CERCA program of the Generalitat de Catalunya. D.B. acknowledges the RCN FRIPRO program for ground-breaking research (catLEGOS project; number 325003) and the RCN program for the National Centers of Excellence (Hylleraas Centre; project number 262695). J.B. and D.B. acknowledge the Norwegian Supercomputing Program (NOTUR; project numbers NN4654K and NS4654K). We would like to thank L. Moran, J. Heras and D. Garay for their invaluable help and support throughout this project in the understanding and adaptation of the AABBA graph kernel, for the design and implementation of the POMs-GibbsNet model respectively and for the review of this manuscript.

Conflict of interest

The authors declare no potential conflict of interests.

Supporting information

Additional supporting information may be found in the online version of this preprint.

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